Anshul Prasashya

+91 9508266641 prasashyaanshul@gmail.com LinkedIn GitHub

Education

Thapar Institute of Engineering and Technology Aug. 2023 – Present
Bachelor of Engineering in Electronics and Computer Engineering

Experience

Amazon ML Summer School 2026 Mentee June 2026 – August 2026
Amazon

- Selected among top 3,000 participants from over 134,000 applicants across India for Amazon ML Summer School 2026.

Machine Learning Research Intern April 2026– Present
Thapar School of Liberal Arts and Sciences *Patiala, Punjab*

- Developed EEG signal preprocessing and feature extraction pipelines utilizing 190+ statistical, spectral, and nonlinear features for behavioral analysis.
- Applied Graph Fourier Transform (GFT)-based EEG analysis to study graph-domain neural representations and functional connectivity patterns.
- Applied EMD and CEEMDAN-based decomposition to improve EEG signal representation, integrating decomposed EEG features with eye-tracking data in a hybrid stacking-ensemble model (Random Forest, Gradient Boosting, XGBoost) for neuromarketing-driven buy/no-buy prediction, achieving Cohen's $\kappa = 0.41$ and F1 = 0.56 under LOOCV — a $\sim$17% improvement in κ over a comparable GFT-based baseline ($\kappa = 0.35$, F1 = 0.54).
- Conducted an IMF-level SHAP ablation identifying mid-range IMFs (5–6) as the dominant contributors to model decisions, guiding feature-set reduction.
- Implemented SHAP-based explainable AI techniques for feature selection and interpretation of neural decision-making behavior.

Projects

RAG Pipeline — Hybrid Retrieval | *React, FastAPI, ChromaDB, Docker, Gemini API* GitHub

- Built a full-stack Retrieval-Augmented Generation (RAG) system with hybrid search, combining BM25 keyword matching and semantic vector search via sentence-transformers and ChromaDB.
- Implemented Reciprocal Rank Fusion (RRF) to algorithmically merge and re-rank results from both retrieval methods, improving relevance over single-method search.
- Designed a decoupled microservices architecture with a React frontend, FastAPI backend, and NGINX reverse proxy, containerized end-to-end with Docker Compose for reproducible deployment.
- Engineered an automated ingestion pipeline (parsing, chunking, embedding, indexing) supporting PDF, DOCX, TXT, CSV, and Markdown, with post-retrieval filtering by source, type, and confidence score.

Facial Expression Recognition | *Python, TensorFlow/Keras, OpenCV, CNN, NumPy* GitHub

- Designed and trained CNN-based deep learning models for real-time facial expression classification using computer vision techniques.
- Developed computer vision preprocessing pipelines using OpenCV, including face detection, normalization, and augmentation for robust facial expression recognition.
- Implemented a real-time computer vision pipeline for facial expression sentiment classification using live webcam streams and static image inputs.

Certifications

- Supervised Machine Learning: Regression and Classification — DeepLearning.AI & Stanford University
- Unsupervised Learning, Recommenders, Reinforcement Learning — DeepLearning.AI & Stanford University
- Getting Started with AI on Jetson Nano — NVIDIA

Technical Skills

Programming Languages: Python, C/C++, JavaScript, HTML/CSS
ML/AI Libraries: TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib
Frameworks & Tools: React.js, Node.js, FastAPI, Docker, ChromaDB, Git, GitHub, VS Code, Jupyter Notebook
Core CS: Data Structures and Algorithms, OOPs, Operating Systems, DBMS, Machine Learning, Deep Learning